4. High-Rise Residential Case Study: Eurovea Tower & Apartment Buildings

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Primary OWL Classes: `ApartmentBuilding`, `EuroveaTower`, `FamousProject`, `ResidentialProject`, `ModernProject`, `Concrete`
Primary OWL Properties: `usesMaterial`, `usesMachine`, `hasSiteManager`, `hasCustomer`, `year`

4.1 Multi-Unit Residential Engineering (Class: ApartmentBuilding)

In the Construction Company Ontology, `ApartmentBuilding` is a direct subclass of `ResidentialProject` (and therefore also a `BuildingProject` and `Project`). Standard catalog instances of `ApartmentBuilding` satisfy the following formal restrictions:

- `year some integer[≥ 2000, < 2026]` (placing them in the `ModernProject` period).
- `usesMachine some ConstructionMachine` (typically tower cranes, concrete pumps, and excavators).
- `usesMaterial some Material` (primarily high-strength reinforced concrete, steel reinforcement, glass facades, and acoustic insulation).

4.2 Landmark Case Study: Eurovea Tower (Class: EuroveaTower)

Eurovea Tower in Bratislava is Slovakia's first skyscraper and the flagship residential reference project in our ontology. In `stavebna_firma_ontology - kópia.owl`, `EuroveaTower` is modeled with multiple inheritance and specific property restrictions:

> **Formal OWL Axioms for `EuroveaTower`:**
>
> - `rdfs:subClassOf FamousProject` (Categorized alongside iconic global landmarks)
> - `rdfs:subClassOf ApartmentBuilding` (Inherits all properties of `ResidentialProject` and `BuildingProject`)
> - `usesMaterial some Concrete` (Ultra-high-strength monolithic reinforced concrete core and slabs)
> - `year value 2023` (`owl:hasValue 2023`, which also entails classification as a `ModernProject` because `2023 ≥ 2000`)

4.3 Technical & Structural Parameters of Eurovea Tower

Technical Parameter	Engineering Specification (Eurovea Tower)
Architectural Height & Floors	168 meters total height | 46 above-ground floors + 3 underground parking levels
Completion Year (`year`)	`2023` (Ontology datatype restriction: `year = 2023`)
Residential Capacity	389 luxury apartments (studio to 5-bedroom penthouses), total floor area 52,400 m²

Primary Structural Material	`Concrete`: 65,000 m³ of C50/60 and C60/75 high-performance concrete; foundation raft slab is 3.0 meters thick poured continuously over 54 hours
Complementary Materials	`Steel` (9,200 tons of B500B reinforcing rebar), `Glass` (triple-glazed wind-resistant curtain wall units), `Insulation` (non-combustible mineral wool A1 fire rating)
Heavy Machinery Deployed (`usesMachine`)	3x Liebherr 280 EC-H self-climbing tower cranes (`Crane`), 4x CAT 336 hydraulic excavators (`Excavator`), 18x 8x4 concrete mixer trucks (`Truck`)
Project Leadership & Crew	`SiteManager`: Ing. Martin Kováč | `Electrician` Lead: Jozef Marek | `Plumber` Lead: Milan Krátky | Client (`Customer`): CUST-101
Total Construction Cost	EUR 115,000,000

4.4 Construction Sequence & Engineering Challenges on Eurovea Tower

Executing a 168-meter `ApartmentBuilding` beside the Danube River required specialized procedures across all four ontology `ConstructionPhase` stages:

1. **`Planning` Phase:** Wind-tunnel aerodynamic modeling in Munich and 3D BIM clash detection for 389 residential plumbing and electrical risers.
2. **`Foundation` Phase:** Secant pile retaining wall around the deep excavation pit to prevent Danube groundwater ingress, followed by drilling 180 large-diameter bored piles (1,200 mm diameter, 28 m depth) beneath the 3.0 m monolithic `Concrete` foundation mat.
3. **`StructuralWork` Phase:** Self-climbing hydraulic formwork system advancing at a pace of one floor every 6 working days using high-early-strength `Concrete` delivered by Danucem (`Supplier`).
4. **`FinishingWork` Phase:** Installation of unitized `Glass` facade elements, pressurized fire-escape stairwell ventilation, smart home electrical metering (by `Electrician` teams), and boosted multi-zone domestic water risers (by `Plumber` teams).